

How the Proof Works

In three-dimensional Navier–Stokes, nonlinear transport can sharpen a flow while viscosity smooths it:

$$\partial_t u + (u \cdot \nabla)u + \nabla p = \nu \Delta u, \quad \nabla \cdot u = 0, \quad u(\cdot, 0) = u_0.$$

The regularity problem is whether that sharpening can keep concentrating a smooth velocity field until it becomes singular in finite time.

A vortex tube gives the physical picture. The surrounding strain pulls it along its axis. Because the fluid is incompressible, the tube cannot gain volume as it lengthens; its cross-section contracts and the rotating core narrows. The strain acts on the vorticity $\omega = \nabla \times u$ through

$$\partial_t \omega + (u \cdot \nabla)\omega = (\omega \cdot \nabla)u + \nu \Delta \omega.$$

Where the vortex is aligned with an amplifying direction of the strain,

$$\omega \cdot ((\omega \cdot \nabla)u) > 0,$$

the stretching part raises $|\omega|$ while the core narrows. Vorticity is one spatial derivative of velocity, so the velocity changes across less distance and its gradients steepen. The nonlinear motion is sharpening the core at the point where viscosity is trying to diffuse it.

Scaling asks how this contest changes when the entire pattern is compressed. For $\lambda > 1$, form

$$u_\lambda(x, t) = \lambda u(\lambda x, \lambda^2 t), \quad p_\lambda(x, t) = \lambda^2 p(\lambda x, \lambda^2 t).$$

A structure of width r , velocity scale U , and duration τ becomes one of width r/λ , velocity scale λU , and duration τ/λ^2 . The characteristic sizes of the nonlinear and viscous terms change as

$$\frac{U^2}{r} \mapsto \lambda^3 \frac{U^2}{r}, \quad \nu \frac{U}{r^2} \mapsto \lambda^3 \nu \frac{U}{r^2}.$$

Direct substitution gives the factor λ^3 in every term of the equation. As the core shrinks, nonlinear sharpening and viscous diffusion strengthen in exact proportion. Scale alone gives neither one an advantage.

Kinetic energy misses this concentration. The compressed core occupies λ^{-3} of the volume while the square of its velocity contributes λ^2 , so

$$\|u_\lambda(\cdot, t)\|_2^2 = \lambda^{-1} \|u(\cdot, \lambda^2 t)\|_2^2.$$

Its gradients can grow while its contribution to the energy falls. Across the homogeneous Sobolev scale,

$$\|u_\lambda(\cdot, t)\|_{\dot{H}^s} = \lambda^{s-\frac{1}{2}} \|u(\cdot, \lambda^2 t)\|_{\dot{H}^s}.$$

Concentration makes the norm smaller below $s = \frac{1}{2}$ and larger above it. At $s = \frac{1}{2}$, it does neither. This is the critical height: the scale at which narrowing the flow gives the measurement no automatic gain and no automatic loss. Write

$$E_s(t) := \frac{1}{2} \|\Lambda^s u(t)\|_2^2, \quad H(t) := E_{1/2}(t).$$

The compressed clock leaves room for a finite-time cascade. If each stage contracts by a fixed factor $0 < \rho < 1$, then

$$r_j = \rho^j r_0, \quad U_j = \rho^{-j} U_0, \quad \tau_j = \rho^{2j} \tau_0,$$

and

$$\sum_{j=0}^{\infty} \tau_j = \tau_0 \sum_{j=0}^{\infty} \rho^{2j} = \frac{\tau_0}{1 - \rho^2} < \infty.$$

Infinitely many narrower and faster stages fit inside a finite interval. This is the Zeno cascade. It does not construct a singular solution. It isolates the difficulty: finite energy does not prevent concentration, and shrinking the scale does not make viscosity overpower the nonlinear term.

Suppose, for contradiction, that the maximal smooth solution ends at $T_* < \infty$.

In order for a singularity to form, velocity has to blow up. In order for that to happen, its acceleration would also have to blow up, which means its jerk has to blow up, and so on, and so on:

$$u, \quad \partial_t u, \quad \partial_t^2 u, \quad \dots$$

What you would expect to see in this pattern, if a singularity were to form, is less spatial regularity as you go up the time derivatives, not more. Navier–Stokes ties each temporal step to a spatial one.

The first three steps are

$$\begin{aligned} \partial_t u &= \nu \Delta u - (u \cdot \nabla)u - \nabla p, \\ \partial_t^2 u &= \nu \Delta \partial_t u - (\partial_t u \cdot \nabla)u - (u \cdot \nabla) \partial_t u - \nabla \partial_t p, \\ \partial_t^3 u &= \nu \Delta \partial_t^2 u - (\partial_t^2 u \cdot \nabla)u - 2(\partial_t u \cdot \nabla) \partial_t u \\ &\quad - (u \cdot \nabla) \partial_t^2 u - \nabla \partial_t^2 p. \end{aligned}$$

The complete pattern is

$$\partial_t^{n+1} u = \nu \Delta \partial_t^n u - \sum_{a=0}^n \binom{n}{a} (\partial_t^a u \cdot \nabla) \partial_t^{n-a} u - \nabla \partial_t^n p.$$

Only now set

$$V_n := \partial_t^n u, \quad Q_n := \partial_t^n p.$$

Every rise in temporal order brings a Laplacian of the preceding response. Viscosity links the time ladder to spatial derivatives. The nonlinear products spread across the lower temporal orders, and pressure differentiates with them, so every row remains the complete equation.

The time ladder and the critical height meet in the exact balance

$$E'_s = -2\nu E_{s+1} + \mathcal{P}_s, \quad \mathcal{P}_s := -\langle \Lambda^s u, \Lambda^s ((u \cdot \nabla)u + \nabla p) \rangle.$$

Repeated differentiation gives

$$H^{(n)} = (-2\nu)^n E_{n+\frac{1}{2}} + \sum_{j=0}^{n-1} (-2\nu)^j \partial_t^{n-1-j} \mathcal{P}_{j+\frac{1}{2}}.$$

The n -th temporal response contains the spatial energy at height $n + \frac{1}{2}$, together with every differentiated nonlinear and pressure transfer. As the temporal order rises, the Sobolev height rises with it.

In three dimensions,

$$s > k + \frac{3}{2} \implies H^s(\mathbb{R}^3) \hookrightarrow C^k(\mathbb{R}^3),$$

so

$$\bigcap_{s < \infty} H^s(\mathbb{R}^3) = H^\infty(\mathbb{R}^3) \hookrightarrow C^\infty(\mathbb{R}^3).$$

A singularity would need the temporal ladder to run toward spatial roughness. If every finite coupled row can be kept on the whole interval, the ladder runs into H^∞ instead.

Take one finite temporal order N and one finite spatial order m . Differentiate only to those orders and keep the resulting velocity, nonlinear, pressure, and viscous terms in one block. The pressure-complete recurrence fixes how its entries overlap; the finite mixed estimate controls the block on the whole interval. In particular,

$$V_N \in L^2(0, T_*; H^{m+1}), \quad V_{N+1} \in L^2(0, T_*; H^{m-1}),$$

and its full norm satisfies

$$\sup_{0 < T < T_*} \mathcal{R}_{N,m}(u; T) < \infty.$$

Viscosity spans the two spatial levels in the adjacent pair; the nonlinear products and pressure remain inside the joined row. The constant may depend on N and m . The proof never asks one constant to control the infinite tower.

Now repeat this finite argument for every N and m . Every block comes from one velocity history generated by u_0 , so entries shared by two blocks already agree. The blocks fit into

$$u_0 \longrightarrow \{\mathcal{R}_{N,m}[u_0]\}_{N,m < \infty} \longrightarrow \mathcal{I}[u_0] := \bigcap_{N,m < \infty} H_t^N(0, T_*; H_x^m).$$

This is where the finite estimates pay off: compatibility turns them into one all-orders intersection without taking an infinite-order limit of their constants.

For each finite m , the intersection contains

$$u \in L^2(0, T_*; H^{m+1}), \quad u_t \in L^2(0, T_*; H^{m-1}).$$

The Hilbert-space trace theorem gives

$$u \in C([0, T_*]; H^m).$$

Because these traces all belong to one velocity field,

$$u_* := u(T_*) \in \bigcap_{m < \infty} H^m = H^\infty.$$

The pressure and velocity recurrences rebuild the complete endpoint jet:

$$Q_N = R_i R_j \sum_{a=0}^N \binom{N}{a} V_{a,i} V_{N-a,j},$$

$$V_{N+1} = \nu \Delta V_N - \sum_{a=0}^N \binom{N}{a} (V_a \cdot \nabla) V_{N-a} - \nabla Q_N.$$

Every V_N and Q_N lies in H^∞ at T_* . Smooth local existence restarts the solution from u_* , and uniqueness glues that restart to the original velocity history. The solution continues past T_* , contradicting maximality. The finite terminal time cannot occur.

This is not an a priori estimate proof. It is an intersection-to-estimate proof. Viscosity relates the velocity's time derivatives to its spatial Sobolev derivatives. Taken through every finite order, those two directions meet in a common intersection. At that intersection, $H^\infty(\mathbb{R}^3) \hookrightarrow C^\infty(\mathbb{R}^3)$ gives smoothness, and the familiar estimates can be taken from whichever finite Sobolev height they require. The estimates come from the intersection, not the other way around.